# AI-Powered Cross-Platform Influencer-Product Suitability Prediction System

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***Abstract****:* Choosing an influencer for a product is still mostly a judgment call based on follower count, perceived popularity, or loose category similarity. This paper develops and evaluates an analytical framework for influencer–product suitability that combines product attributes, influencer characteristics, campaign context, platform information, and historical campaign outcomes, while keeping pre-campaign predictors strictly apart from post-campaign outcomes. The empirical study uses a joined multi-table dataset of 100,000 orders, 60 influencer–campaign performance records, 90,000 customer records, 25 products, 20 influencers, and 10 campaigns. After validation and aggregation, the main analytical table holds 312 influencer–campaign–product observations. A 35-field leakage audit separates 14 pre-campaign variables, 9 historical outcome variables, 7 constructed score or target fields, 3 identifiers, and 2 campaign metadata fields that need availability checks. Under a temporal split, a random-forest baseline reaches an MAE of 233,068, an RMSE of 318,326, and an $R^2$ of 0.920 on a 60-row held-out set, while a mean predictor reaches an $R^2$ of −0.075. In a separate pre-campaign comparison the random forest obtains an $R^2$ of 0.778 and a Spearman ρ of 0.904, and a hand-specified suitability score performs far worse, which shows that a transparent score and a predictive model answer different questions. In ranking experiments, a revised score built from follower count and product price, with weights chosen on training-period data, reaches $\rho = 0.858$ and a top-10% revenue lift of 2.634 on the evaluated test set, whereas audience match alone is not a reliable ranking signal in this dataset. The results favor leakage-aware machine learning for suitability ranking, but they come from a small, project-supplied dataset and do not establish causal effects on sales.

Keywords: Influencer marketing, influencer–product fit, recommender systems, machine learning, suitability ranking, cross-platform comparison, feature leakage, explainable AI, digital marketing.

## I. INTRODUCTION

Brands now lean on creators to carry products to audiences, yet the research does not treat influencer effectiveness as one quantity. Product–influencer fit, credibility, follower characteristics, content value, platform context, and influencer size can all shape the outcome. Belanche et al. showed that congruence among the influencer, the product, and the consumer can change attitudes and behavioral intentions [1], and Janssen et al. found that product–influencer fit and follower count interact instead of acting independently [2].

For a decision-support system this creates a practical difficulty. A brand does not simply need a list of popular influencers. It needs a ranking that can take in the product category, product traits, campaign timing, platform, audience fit, and influencer scale, and that uses only the information actually known before the campaign starts. In practice several things make the choice hard: a category mismatch can undermine a recommendation even when the influencer is large; follower count is easy to observe but says little about content relevance, audience quality, or campaign context; engagement and reach are hard to compare across platforms; season and timing matter for products whose demand shifts; and a ranking that cannot be explained is difficult to audit or defend in a business setting.

The behavioral literature has studied many of these factors one at a time, whereas computational matching systems are less developed. Sweet et al. demonstrated machine-learning brand–influencer matchmaking based on content similarity [3], and Weinlich and Semerádová tie congruence measurement directly to recommender-system methods [4]. Our question is a more specific one: whether heterogeneous influencer, product, and campaign variables can be combined into a reproducible suitability-ranking framework and evaluated without letting post-campaign outcomes leak into pre-campaign features.

We therefore treat influencer–product suitability as an analytical ranking problem, and the scope is deliberately narrower than causal sales prediction. Historical revenue and order counts help us check whether a ranking is associated with outcomes, but they cannot serve as inputs to a pre-campaign model, because those outcomes occur after the decision point. This distinction drives the data audit, the temporal split, the feature engineering, and the way results are interpreted.

The paper makes three contributions. First, it documents a feature-governance procedure, a 35-field leakage audit, that decides which variables may enter a pre-campaign model. Second, it compares a transparent suitability score with a leakage-aware random forest on held-out data. Third, it evaluates several ranking methods, including popularity and audience-match baselines, a revised score, and alternative temporal cutoffs. Section II reviews related work, Section III states the objectives and hypotheses, and Section IV presents the framework. Sections V to VII describe the data, preprocessing, and evaluation design, Section VIII reports results, and Sections IX to XIII cover discussion, architecture, ethics, limitations, and conclusions.

## II. RELATED WORK

### *A. Influencer Effectiveness and Credibility*

A large share of the behavioral work asks why influencer content persuades. Lou and Yuan linked informative value, trustworthiness, attractiveness, and similarity to trust in branded influencer content and its downstream outcomes [5]. Saima and Khan found that trustworthiness, information quality, and entertainment value shape credibility and, through it, purchase intention [6], while Coutinho et al. examined expertise, trustworthiness, and attractiveness as credibility dimensions tied to purchase intention [7]. Dauhan and Langi, working in a stimulus–organism–response framework, placed credibility and content value upstream of attitude, online engagement, and purchase intention [8]. Foroughi et al. used PLS and fsQCA to show that several different combinations of credibility, content, attractiveness, and attitude can lead to high purchase intention [9], which argues against selection rules built on a single factor. A recent review of source credibility theory points the same way: attractiveness, expertise, and trustworthiness shape perceived credibility, and influencer–brand fit and disclosure transparency matter as conditions [10]. Leung et al. examined sender, receiver, and message factors and reported nonlinear moderating effects, which cautions against one-dimensional influencer selection [11].

Several syntheses now organize this evidence. Vrontis et al. reviewed 68 articles, found fragmented research streams, and called for integrative work across antecedents, mediators, moderators, and contexts [12]. Joshi et al. reviewed 214 articles and grouped the factors into audience, brand, content, influencer, social, and technology categories, with authenticity, sponsorship, and engagement as major streams [13]. The meta-analysis by Ao et al., covering 62 studies and 22,554 participants, related entertainment, credibility, congruence, expertise, trustworthiness, and other attributes to engagement or purchase intention [14]. Pan et al. analyzed 1,531 effect sizes from 251 papers, grouped determinants into post, follower, and influencer characteristics, and showed that effects depend on platform and product type [15]. Barari et al. synthesized 71 papers and 135 experiments and found that influencer size interacts with other conditions: smaller influencers can be stronger for engagement, whereas larger ones can be stronger for purchase intention [16].

### *B. Congruence and Fit*

Fit between influencer and product is a recurring theme. Belanche et al. showed that congruence across influencer, product, and consumer can influence product attitudes, purchase intentions, and recommendation intentions [1]. In an experiment, Janssen et al. found that product–influencer fit and follower size work together, with credibility and identification mediating the effects [2]. Mettenheim and Wiedmann separated influencer–brand fit from influencer–consumer self-congruence and showed that different congruence mechanisms matter under different involvement conditions [17]. Dhun and Dangi connected expertise, similarity, trustworthiness, and congruence to brand attitude and electronic word-of-mouth intentions [18], and Chopra et al. identified trust, personal relevance, inspiration, and product–influencer fit as relevant to consumer behavior, stressing category-specific influencer choice [19].

### *C. Computational Matching and Recommender Systems*

The computational side is thinner. Sweet et al. demonstrated a machine-learning approach to brand–influencer matchmaking built on content similarity [3]. Weinlich and Semerádová operationalized influencer–brand personality congruence for recommender systems and compared similarity measures; Euclidean distance performed best in their tested setup [4]. Bawack and Bonhoure classified 52 variables into 12 categories and argued that influencer research can inform theory-driven social recommender systems [20]. Liu combined credibility, parasocial relationships, and recommendation-algorithm context in a model of purchase intention, which links influencer marketing to smart recommendation environments [21].

### *D. AI and Virtual Influencers*

Virtual and AI influencers add a further dimension. Laszkiewicz and Kalińska-Kula reviewed 35 articles and describe a growing but still developing research stream [22]. Sands et al. found similar personalization and following intention for AI and human influencers but lower source trust for AI influencers [23], which suggests that influencer type should be treated as context and not assumed equivalent. Franke et al. reported that virtual influencers can raise novelty, although perceived congruence with the product category still matters for effectiveness [24]. Jayasingh et al. found credibility, informative value, and human-likeness to be major drivers of engagement and purchase intention in an AI-influencer setting [25].

### *E. Platform Behavior, Text Analytics, and Institutional Evidence*

Work from the authors' own institute adds platform, text-analytics, and governance context that bears on this problem. Yalparatte and Pawar compared engagement before and after the 2023 TikTok usage-time restrictions and documented a shift of usage and engagement growth toward alternative platforms [26]. This supports treating platform context as something that changes over time, and it reinforces why engagement is hard to compare across platforms. Brahme et al. addressed forecasting of social-networking-site usage in an education setting [27], which points to the same need for temporal treatment of platform-level trends. On the text side, Mandhane and Wadkar compared TF-IDF-based classical models (Naive Bayes and a linear SVM), a lexicon baseline, and a BERT-based model for sentiment classification of Amazon product reviews [28], and Wadkar and Garud examined NLP approaches for processing social-media text [29]. Both are relevant to the text-based category matching proposed here as future work. Patil studied the performance of e-marketing customers in Pune [30], which gives a regional digital-marketing reference point. Finally, Kamble et al. proposed a conceptual framework for agentic AI that defines roles, operational boundaries, and trust mechanisms [31], and Pawar and Bandgar applied explainable AI to credit scoring on Open Banking data [32]; both bear on the governance and explainability requirements discussed in Sections IX and XI. Most of this work is descriptive,

conceptual, or review-based, and none of it ranks influencers, so it supplies context and methodological precedent and not a competing method.

*F. Synthesis and Research Gap*

Six factor groups recur across these studies: influencer attributes, audience characteristics, product attributes, content characteristics, platform or channel context, and campaign conditions. The literature therefore supports a multi-dimensional view of selection. At the same time, much of the evidence comes from surveys, experiments, structural equation models, or meta-analyses of consumer outcomes, not from reproducible predictive pipelines built on multi-table campaign data. Computational studies exist, but their objectives and variables differ from those in the present dataset.

The gap we address is partly methodological. Influencer research mostly studies consumer response, not pre-campaign computational ranking. Fit is usually treated as a psychological construct, and fewer studies turn it into a reproducible multi-factor ranking problem. Cross-platform normalization matters conceptually but is hard in practice because platform metrics are not directly comparable, and platform usage itself shifts over time, as the TikTok-restriction study shows [26]. Historical outcome variables can leak into predictors when they are used for pre-campaign decisions. And explainability is usually discussed in conceptual terms, whereas here explanation is an output that should be traceable to input factors. The field needs workflows that make feature timing explicit, compare heuristic scores with leakage-aware predictive models, and report ranking quality alongside self-reported attitudes.

## III. RESEARCH OBJECTIVES, QUESTIONS, AND HYPOTHESES

The study pursues seven objectives:

**O1.** Define a reproducible influencer–product suitability framework from product, influencer, platform, campaign, and temporal factors.

**O2.** Construct and validate a multi-table analytical dataset of influencer–campaign–product observations.

**O3.** Audit feature availability and leakage before model training.

**O4.** Compare a transparent suitability score with supervised machine-learning baselines.

**O5.** Evaluate ranking quality with Spearman correlation and top-decile revenue lift.

**O6.** Test whether ranking performance holds across alternative temporal cutoffs.

**O7.** Provide an interpretable path from feature values to ranking decisions.

These objectives translate into five research questions:

**RQ1.** Can influencer–product suitability be represented with observable pre-campaign variables?

**RQ2.** How much does a leakage-aware random forest improve over a naive mean predictor?

**RQ3.** How does a transparent suitability score compare with the supervised model?

**RQ4.** Which ranking signals are strongest in the current dataset?

**RQ5.** Does ranking performance stay positive across different temporal cutoffs?

and four hypotheses:

**H1.** A supervised model using pre-campaign features will outperform a mean baseline on held-out outcome prediction.

**H2.** A multi-factor suitability ranking will show a positive association with held-out campaign outcomes.

**H3.** A model using leakage-safe pre-campaign variables will outperform a historical-outcome-based heuristic when the task is framed as prospective prediction.

**H4.** Ranking performance will vary across temporal cutoffs, reflecting differences in campaign mix and sample size.

## IV. CONCEPTUAL FRAMEWORK

The approved conceptual design names four intended dimensions: category match, normalized engagement, seasonal relevance, and platform fit. The empirical implementation broadens these into measurable fields, because the supplied dataset contains explicit product, influencer, campaign, and temporal variables. The key methodological choice is that the framework serves as a decision-support structure, while predictive validation uses only features that are available at decision time. Fig. 1 shows the analytical pipeline, and Table I lists the factor groups used in the operational framework.

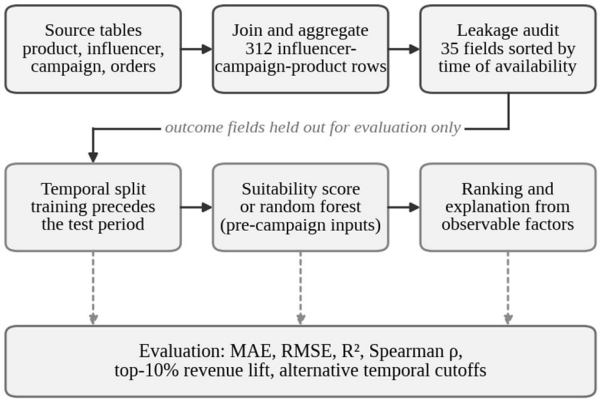



Fig. 1. Proposed analytical pipeline, from product, influencer, and campaign inputs to ranking and explanation.

TABLE I
OPERATIONAL FACTORS IN THE STUDY

| Factor group | Operational variables | Purpose |
|---|---|---|
| Product | Category, sub-category, season, list price | Product identity and demand context |
| Influencer | Platform, follower count, tier, audience-location match | Reach, scale, and audience fit |
| Campaign | Platform, goal, budget, start date, end date, duration | Campaign context and timing |
| Temporal | Campaign start and end, cutoff date | Prospective train/test boundary |
| Outcome references | Order count, revenue, conversions, ROI | Evaluation only; excluded from predictive features |
| Explainability | Feature contribution or rationale | Human-readable basis for ranking |

## V. DATASET AND DATA PREPARATION

### *A. Source Data*

The empirical study uses the Power BI–style marketing campaign dataset supplied with the project. Its source tables cover orders, influencer campaign performance, customer demographics, products, influencers, and campaigns (Table II and Fig. 2). The project archive also contains alternative supplementary datasets, but the experiments reported here rely on the six-table source set, because it links influencer, campaign, product, and outcome fields. Fig. 3 shows how the entities relate to one another.

TABLE II
SOURCE TABLE DIMENSIONS

| Source table | Rows | Columns |
|---|---|---|
| influencers.xlsx | 20 | 9 |
| influencer campaign performance.xlsx | 60 | 8 |
| customer demographics.xlsx | 90,000 | 11 |
| orders.xlsx | 100,000 | 12 |
| products.xlsx | 25 | 7 |
| campaigns.xlsx | 10 | 10 |

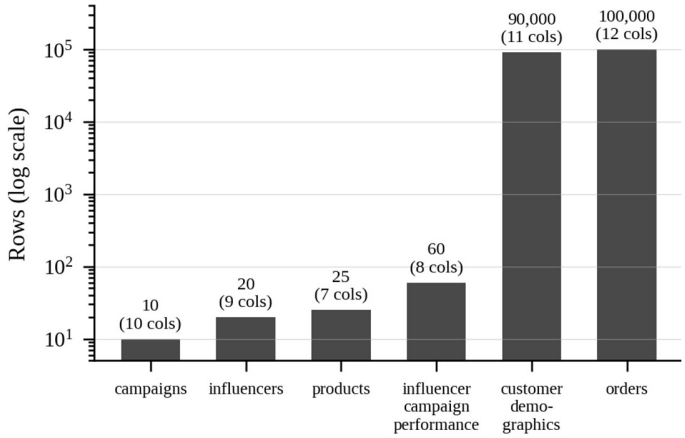



Fig. 2. Size of the six source tables. A logarithmic axis is used because row counts differ by several orders of magnitude.

### *B. Validation and Data Quality*

After joining, 100,000 order-level rows were retained and then aggregated into 312 influencer–campaign–product rows. The aggregated table covers 16 distinct influencers, 10 campaigns, and 25 products. No duplicate Order_ID values appear at the order level, no duplicate keys appear in the aggregated table, and there are no product-category mismatches (Table III). Two quality issues remain. The aggregated table has 216 missing Main_Product_ID values, and 73,219 order/campaign platform mismatches were found. We keep the mismatches as an observed data-quality issue and do not relabel them, because the source systems record different channel fields.

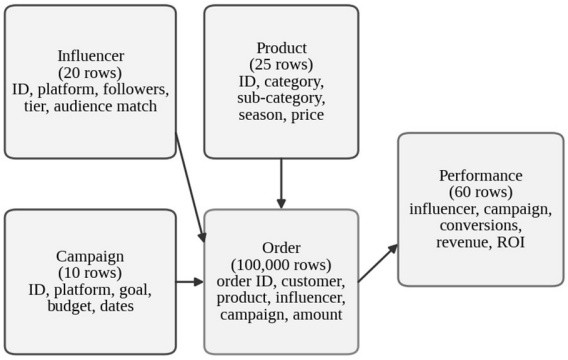
Campaign dates are complete. Start dates run from 2024-05-08 to 2025-05-12 and end dates from 2024-06-07 to 2025-06-28, and no row is missing either date. These dates provide the basis for temporal splitting. Numeric summaries of the 312-row analytical table are given in Table IV.

TABLE III
DATA-VALIDATION SUMMARY

| Check | Result |
|---|---|
| Order-level joined rows | 100,000 |
| Aggregated rows | 312 |
| Order-level duplicate Order_ID | 0 |
| Aggregated duplicate key rows | 0 |
| Missing values in aggregated table | 216 |
| Campaign-platform mismatches | 73,219 |
| Product-category mismatches | 0 |

TABLE IV
NUMERIC DESCRIPTIVE SUMMARY OF THE 312-ROW ANALYTICAL DATASET

| Variable | N | Mean | Std. dev. | Minimum | Median | Maximum |
|---|---|---|---|---|---|---|
| Product_List_Price | 312 | 3,429.21 | 2,032.17 | 608.00 | 3,127.00 | 8,526.00 |
| Influencer_Follower_Count | 312 | 677,981.49 | 618,818.93 | 6,036.00 | 326,621.00 | 1,888,335.00 |
| Audience_Location_Match_pct | 312 | 65.51 | 9.93 | 54.20 | 61.00 | 86.40 |
| Campaign_Budget | 312 | 787,616.25 | 435,679.89 | 152,109.00 | 725,044.00 | 1,396,414.00 |
| Campaign_Duration_Days | 312 | 36.02 | 13.13 | 22.00 | 29.00 | 60.00 |
| Order_Count | 312 | 320.51 | 264.16 | 36.00 | 212.00 | 1,581.00 |
| Total_Quantity | 312 | 365.79 | 302.10 | 39.00 | 242.00 | 1,800.00 |
| Total_Revenue | 312 | 1,130,932.36 | 1,207,938.87 | 31,571.70 | 732,590.80 | 9,779,291.90 |
| Total_Discount | 312 | 104,609.46 | 137,170.79 | 858.30 | 54,077.04 | 1,085,508.10 |
| Unique_Customers | 312 | 320.34 | 263.89 | 36.00 | 212.00 | 1,577.00 |
| Performance_Conversions | 312 | 1,666.75 | 1,103.94 | 290.00 | 1,135.00 | 4,724.00 |
| Performance_Revenue | 312 | 5,991,106.76 | 4,116,476.34 | 1,148,902.66 | 4,628,991.47 | 15,254,159.96 |
| Performance_Cost | 312 | 48,673.08 | 46,853.50 | 900.00 | 24,000.00 | 120,000.00 |
| Performance_ROI | 312 | 279.92 | 334.59 | 57.24 | 163.79 | 1,439.26 |




*The customer table (90,000 rows) gives context for aggregation only.*

Fig. 3. Analytical entity relationships used to build the modelling table

## VI. PREPROCESSING AND LEAKAGE AUDIT

### *A. Preprocessing*

Preprocessing joined the entity tables through stable identifiers, standardized the dates, and aggregated order-level measures to the influencer–campaign–product level. Campaign dates were converted into date and duration features that could have been known before launch. Outcome variables were kept as separate evaluation fields and excluded from every predictive feature matrix. Each field was then audited for temporal availability and assigned to one of five classes, and a temporal cutoff ensured that training observations precede the held-out test period. These steps make the decision boundary explicit: variables observed after the campaign are not allowed into a prospective predictor.

### *B. Leakage Audit and Feature Set*

Table V summarizes the audit of 35 fields, and Fig. 4 shows the class sizes. Fourteen fields qualify as pre-campaign features and form the predictive feature set. Product_Focus_Type and Main_Product_ID need explicit availability verification before they can be treated as safe predictors; Main_Product_ID is also missing in 216 aggregated rows. Historical outcome fields such as Order_Count, Total_Revenue, Performance_Revenue, and Performance_ROI are excluded from all prospective models.

TABLE V
FEATURE LEAKAGE AUDIT BY FIELD CLASS

| Class | n | Fields | Handling |
|---|---|---|---|
| Pre-campaign features | 14 | Product_Category, Product_Sub_Category, Product_Season, Product_List_Price, Influencer_Platform, Influencer_Follower_Count, Influencer_Tier, Audience_Location_Match_pct, Campaign_Platform, Campaign_Goal_Type, Campaign_Budget, Campaign_Start_Date, Campaign_End_Date, Campaign_Duration_Days | Used as categorical or numeric features; date features derived; duration only if known before launch |
| Historical outcomes | 9 | Order_Count, Total_Quantity, Total_Revenue, Total_Discount, Unique_Customers, Performance_Conversions, Performance_Revenue, Performance_Cost, Performance_ROI | Excluded from predictive features |
| Constructed score or target | 7 | Engagement, Revenue, Audience_Match, Platform, and Season components; Suitability_Score; Suitability_Rank | Descriptive only; never model inputs |
| Identifiers | 3 | Source_Influencer_ID, Campaign_ID, Product_ID | Excluded from numeric modeling; kept for grouping, splitting, ranking |
| Campaign metadata | 2 | Product_Focus_Type, Main_Product_ID | Use only if known before the campaign; otherwise exclude |

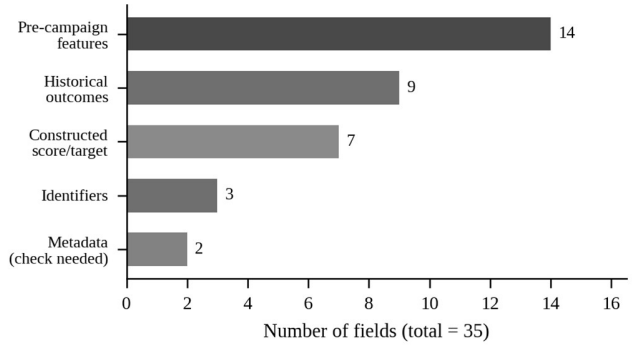



Fig. 4. Classification of the 35 audited fields by temporal availability.

## VII. MODELS AND EVALUATION DESIGN

### *A. Supervised Experiments*

Two related questions are evaluated separately. The principal baseline experiment measures direct outcome prediction on a fixed partition of 252 training rows and 60 test rows, split at the cutoff date 2025-03-24. A dedicated pre-campaign comparison then asks whether a leakage-safe random forest predicts held-out outcomes better than a manually specified suitability score. The two experiments came from different evaluation pipelines, so we report them side by side and do not merge them into a single headline statistic. Both compare the random forest with a dummy mean baseline using MAE, RMSE, and $R^2$. The target is a continuous held-out campaign outcome whose errors are on the scale of revenue, but the retained experiment records do not name the target field Category Match Score, Engagement Quality Score, Seasonal Relevance Score, Platform Fit Score Suitability Score, Suitability Percentage, Suitability Rank, Recommendation level, Hyperparameters and encoding details are also not recorded in the retained materials.

### *B. Suitability Scores*

The original project specification defines a transparent score from category match, normalized engagement, seasonal relevance, and platform fit:

$$S = 0.40C + 0.30E + 0.20R + 0.10P \quad (1)$$

where $S$ is the suitability score and $C$, $E$, $R$, and $P$ are the category-match, engagement-quality, seasonal-relevance, and platform-fit scores. The reported experiments do not evaluate this four-component formula as such. They use operational scores built from the fields available in the dataset.

The descriptive score artifact combines min-max normalized order count and revenue with audience-location match, platform match, and season availability (Table VI). Because

order count and revenue are historical outcomes, this artifact serves as a descriptive benchmark and is not a leakage-safe predictor. The pre-campaign suitability score v1 instead weights audience match, follower count, campaign budget, and product price at 0.35, 0.25, 0.25, and 0.15. Its Spearman ρ of 0.488 matches that of the original score in the revision experiment, so we treat the two as the same score.

TABLE VI
SPECIFICATION OF THE DESCRIPTIVE SCORE ARTIFACT

| Field | Calculation | Weight |
|---|---|---|
| Engagement_Component | Min-max normalized Order_Count | 30% |
| Revenue_Component | Min-max normalized Total_Revenue | 25% |
| Audience_Match_Component | Min-max normalized Audience_Location_Match_pct | 20% |
| Platform_Component | 1 if influencer and campaign platforms match, else 0 | 15% |
| Season_Component | 1 when product season is available, else 0 | 10% |
| Suitability_Score | Weighted sum of components × 100 | — |
| Suitability_Rank | Dense descending rank of Suitability_Score | — |

### *C. Ranking Evaluation*

Ranking quality is measured with Spearman's ρ between a score and the observed outcome, NDCG at k, and top-10% revenue lift. Lift is the mean revenue of the top-ranked decile divided by the overall mean revenue, so values above 1 mean that the selected decile beats the average. Three ranking methods are compared: the proposed suitability ranking, a popularity-only baseline AI-Powered Cross-Platform Influencer-Product Suitability Prediction System, Popularity Measure should represent the overall audience size and content reach of an influencer. and an audience-match-only ranking. A retrospective revenue reference ranks by the observed outcome itself. It is an upper-bound benchmark and not a deployable method.

### *D. Temporal Robustness Checks*

To test stability, the ranking was re-evaluated at four alternative cutoff dates: 2024-11-01, 2025-01-01, 2025-03-24, and 2025-05-01. Each cutoff produces its own partition. These partitions were generated separately from the principal split, so their row counts should not be compared with it.

### *E. Reporting Conventions*

Where a metric is not defined for a method, such as MAE for a pure ranking comparison or Spearman's ρ for a constant dummy predictor, the tables read "N/A." Every other value is reported as stored in the experiment output, and no value has been imputed or replaced to fill a cell.

## VIII. RESULTS

### *A. Principal Supervised Baseline*

Table VII reports the principal baseline, trained on 252 rows and evaluated on 60. The random forest reaches an MAE of 233,068, an RMSE of 318,326, and an $R^2$ of 0.920. The dummy mean baseline reaches an MAE of 785,902, an RMSE of 1,165,398, and an $R^2$ of −0.075. Fig. 5 contrasts the two error measures.

TABLE VII
PRINCIPAL BASELINE EXPERIMENT

| Model | Train | Test | MAE | RMSE | $R^2$ |
|---|---|---|---|---|---|
| Dummy mean baseline | 252 | 60 | 785,902 | 1,165,398 | −0.075 |
| Random forest baseline | 252 | 60 | 233,068 | 318,326 | 0.920 |

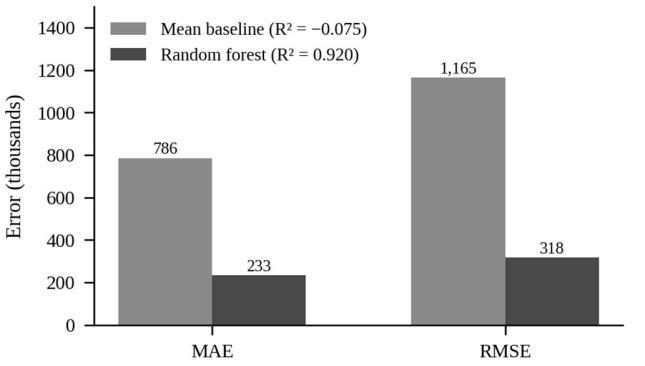



Fig. 5. Error of the mean baseline and the random forest in the principal experiment (60-row test set).

### *B. Pre-Campaign Model Comparison*

Table VIII and Fig. 6 give the dedicated pre-campaign comparison. The random forest reaches an MAE of 355,317, an RMSE of 510,646, an $R^2$ of 0.778, and a Spearman ρ of 0.904. The pre-campaign suitability score v1 reaches an MAE of 1,225,813, an RMSE of 1,635,582, an $R^2$ of −1.282, and a ρ of 0.488. The dummy baseline has an $R^2$ of −0.017.

TABLE VIII
DEDICATED PRE-CAMPAIGN MODEL COMPARISON

| Model | MAE | RMSE | $R^2$ | Spearman ρ |
|---|---|---|---|---|
| Dummy mean baseline | 778,004 | 1,091,974 | −0.017 | N/A |
| Random forest baseline | 355,317 | 510,646 | 0.778 | 0.904 |
| Pre-campaign suitability score v1 | 1,225,813 | 1,635,582 | −1.282 | 0.488 |

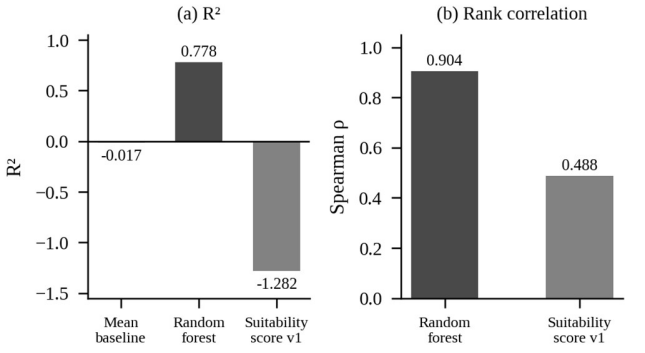



Fig. 6. Pre-campaign comparison: (a) $R^2$ and (b) Spearman ρ for the random forest and the suitability score v1.

### *C. Descriptive Suitability Scores*

The descriptive suitability score, computed for all 312 rows with no missing values, ranges from 12.1756 to 79.5963, with

a mean of 29.3575 and a median of 29.3651. In the 60-row evaluation set the score shows positive Spearman correlations with Total_Revenue ($\rho = 0.423$) and Order_Count ($\rho = 0.447$), and the observed outcomes correlate with each other at $\rho = 0.782$ (Table IX).

TABLE IX
DESCRIPTIVE RANKING EVALUATION (60-ROW EVALUATION SET)

| Comparison | n | ρ | p-value | NDCG@k |
|---|---|---|---|---|
| Score vs. Total_Revenue | 60 | 0.423 | $7.66\times10^{-4}$ | 0.888 |
| Score vs. Order_Count | 60 | 0.447 | $3.42\times10^{-4}$ | 0.910 |
| Revenue vs. Order_Count (observed) | 60 | 0.782 | $1.71\times10^{-13}$ | N/A |

### *D. Ranking-Method Comparison*

Table X and Fig. 7 compare the ranking methods. The proposed suitability ranking reaches $\rho = 0.455$ and a top-10% lift of 1.425. The popularity-only baseline is stronger, with $\rho = 0.551$ and a lift of 1.997. Audience-match-only ranking performs poorly, with $\rho = -0.277$ and a lift of 0.607. The retrospective revenue reference, which has access to the observed outcome, reaches $\rho = 1$ and a lift of 3.063.

TABLE X
RANKING-METHOD COMPARISON

| Method | ρ | p-value | Top-10% mean revenue | Lift |
|---|---|---|---|---|
| Proposed suitability | 0.455 | $1.56\times10^{-6}$ | 1,746,450 | 1.425 |
| Popularity-only | 0.551 | $2.01\times10^{-9}$ | 2,447,889 | 1.997 |
| Audience-match-only | −0.277 | $4.82\times10^{-3}$ | 744,187 | 0.607 |
| Retrospective revenue reference | 1.000 | ≈ 0 | 3,755,025 | 3.063 |

Overall mean revenue is 1,225,848 for all methods.

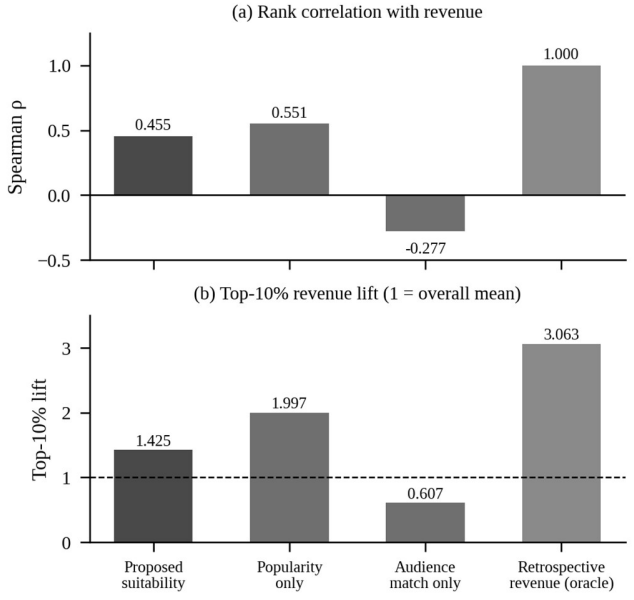



Fig. 7. Ranking-method comparison: (a) Spearman ρ with revenue and (b) top-10% revenue lift. The retrospective reference uses observed outcomes and is shown only as an upper bound.

### *E. Revised Score*

The revised score (Table XI and Fig. 8) puts a weight of 0.40 on follower count and 0.60 on product price, and gives audience match and campaign budget zero weight. These weights were selected using training-period Spearman ranking correlation, whereas the original weights were specified manually. On the evaluated test set the revised score reaches $\rho = 0.858$ and a top-10% lift of 2.634, compared with $\rho = 0.488$ and a lift of 1.700 for the original score.

TABLE XI
ORIGINAL VERSUS REVISED SCORE

| Score | Weights (aud./foll./budget/price) | ρ | p-value | Top-10% mean revenue | Lift |
|---|---|---|---|---|---|
| Original v1 | 0.35/0.25/0.25/0.15 | 0.488 | $2.00\times10^{-7}$ | 2,084,395 | 1.700 |
| Revised v1 | 0/0.40/0/0.60 | 0.858 | $1.12\times10^{-30}$ | 3,229,422 | 2.634 |

Overall mean revenue is 1,225,848. Original weights: manually specified. Revised weights: training-period Spearman ranking correlation.

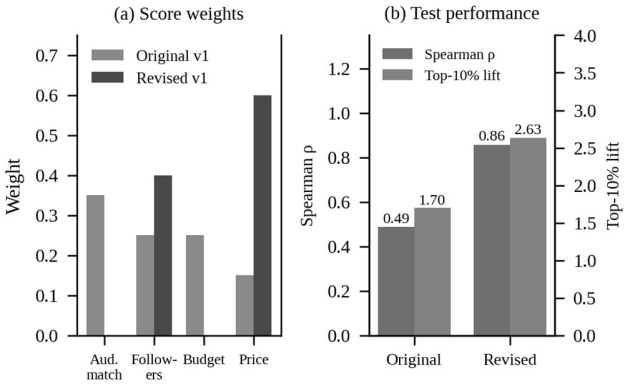



Fig. 8. Original versus revised score: (a) component weights and (b) test-set Spearman ρ and top-10% lift.

### *F. Temporal Robustness*

Spearman correlations are positive at all four cutoffs, from 0.211 to 0.872, and top-10% lifts range from 1.364 to 2.052 (Table XII and Fig. 9). The highest correlation occurs at the latest cutoff, where only one test campaign remains.

TABLE XII
ALTERNATIVE TEMPORAL CUTOFFS

| Cutoff | Train / test | Test camp. | ρ | p-value | Top-10% / all mean (k) | Lift |
|---|---|---|---|---|---|---|
| 2024-11-01 | 120 / 192 | 6 | 0.211 | $3.27\times10^{-3}$ | 1,601 / 1,174 | 1.364 |
| 2025-01-01 | 192 / 120 | 4 | 0.560 | $2.88\times10^{-11}$ | 2,356 / 1,307 | 1.803 |
| 2025-03-24 | 210 / 102 | 3 | 0.524 | $1.54\times10^{-8}$ | 2,515 / 1,226 | 2.052 |
| 2025-05-01 | 282 / 30 | 1 | 0.872 | $3.65\times10^{-10}$ | 2,473 / 1,361 | 1.818 |

Mean revenue columns are in thousands.

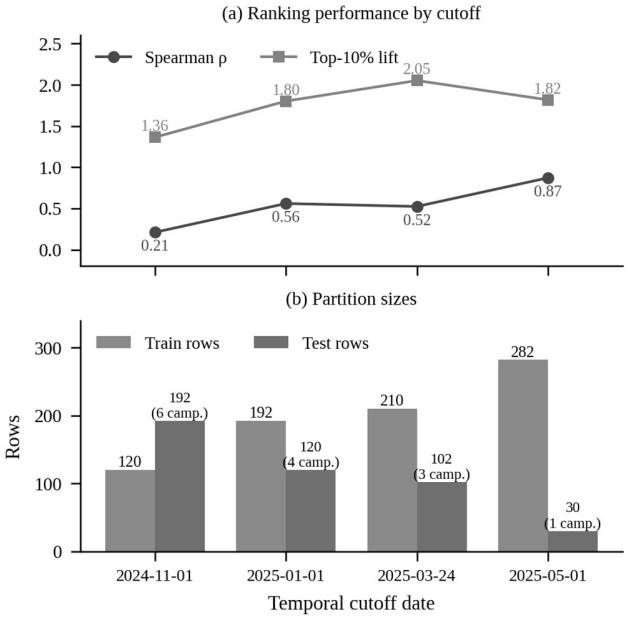



Fig. 9. Robustness across temporal cutoffs: (a) Spearman ρ and top-10% lift; (b) training and test partition sizes, with test campaign counts.

## IX. DISCUSSION

### A. Summary of Findings

Several findings stand out. The multi-table source reduces to a usable 312-row analytical table, and its data-quality problems (216 missing Main_Product_ID values and 73,219 platform mismatches) stay visible rather than hidden. The field-by-field audit mattered, because historical revenue, order count, conversions, and ROI are not valid pre-campaign predictors. The random forest clearly beat the mean baseline in both supervised experiments, although the size of the gain differs between pipelines. A transparent score built on historical outcomes showed positive descriptive ranking association but poor prospective regression performance. Popularity-only ranking outperformed the original multi-factor ranking in one held-out comparison, which shows that adding factors does not automatically improve results, and audience-location match alone performed poorly. The revised score ranked strongly, but its weights are dataset-specific, and robustness was positive yet variable across cutoffs.

### B. Hypotheses

Table XIII relates the results to the four hypotheses.

TABLE XIII
OUTCOME OF THE HYPOTHESES

| Hypothesis | Outcome | Supporting result |
|---|---|---|
| H1: model beats mean baseline | Supported | $R^2$ 0.920 vs. −0.075 (principal); 0.778 vs. −0.017 (pre-campaign) |
| H2: positive ranking association | Supported | ρ = 0.455 (proposed), 0.488 (original v1), 0.858 (revised); popularity-only is higher (0.551) than proposed |
| H3: leakage-safe model beats heuristic | Supported against score v1 | $R^2$ 0.778 vs. −1.282; the historical-outcome score was not part of this regression comparison |
| H4: performance varies by cutoff | Supported | ρ from 0.211 to 0.872; lift from 1.364 to 2.052 |

### C. Interpretation

The results are consistent with the broad view that influencer effectiveness is multidimensional, with credibility, attractiveness, congruence, audience characteristics, content, and platform context all mattering [1], [5], [14]–[17]. The strong showing of follower count in the revised score fits the finding that influencer size interacts with other conditions [16], and it suggests that size stays an informative signal in this dataset. The experiments also add a practical machine-learning point: once a real prediction task is defined, a feature's usefulness depends on when it can be observed, not only on whether it correlates with historical outcomes.

The gap between the heuristic score and the random forest is the most telling result. A transparent score is easy to explain and audit, but it may miss nonlinear relationships and interactions. The random forest can capture interactions among follower count, price, category, platform, and campaign variables, which may explain its higher accuracy here; we did not test this directly. The popularity-only result is a useful warning in the other direction: a more complicated ranking formula is not automatically better than a simple benchmark.

The revised score should not be read as showing that price and followers are the most important selection variables in general. They gave the strongest ranking signal under one specific training-period selection procedure. Revenue is partly a function of product price, so price may track the outcome almost mechanically, although this possibility was not tested. The cutoff analysis supports a cautious reading as well. Performance varies materially, and the latest cutoff has a single test campaign, so its correlation of 0.872 should not carry much weight. The variation is consistent with a dataset that has few campaigns.

Recommender-system research is also relevant. Bawack and Bonhoure position influencers as social recommenders [20], and Weinlich and Semerádová connect congruence modeling with recommender similarity measures [4]. The present framework differs by focusing on a campaign-level suitability decision and by enforcing explicit leakage controls. That makes the workflow less flashy than a black-box recommendation claim, but easier to defend empirically.

### D. Practical Implications

For practitioners, the lesson is operational. Influencer selection should start from a clear decision date and a list of variables known at that date. Historical performance should be used to train and validate models, but historical outcomes should not quietly re-enter the feature matrix. Simple popularity baselines are worth keeping, since they give an honest benchmark that more complex systems must beat. For researchers, the study offers a reproducible pattern for combining behavioral literature with empirical data engineering. A paper on influencer matching should report which fields were actually available, how they were joined, what was excluded, how the temporal split was defined, and how ranking quality changed across baselines.

### E. Explainability and Traceability

Explainability here is a reporting layer and not a claim about a specific proprietary XAI algorithm. For each ranked

candidate, the system should expose the observable factors behind the score or model decision, such as follower scale, product price, category, platform, campaign goal, and temporal context. A future implementation could use SHAP or another model-agnostic method to produce local contribution values for the random forest. Explainable-AI methods have already been applied to scoring models built on API-sourced transaction data [32], which supports this direction. No SHAP results were produced in the completed experiments.

## X. SYSTEM ARCHITECTURE

Fig. 10 shows a conceptual architecture for implementation. The input layer takes product, influencer, and campaign data. The validation layer runs schema, missingness, foreign-key, and field-availability checks. The feature layer handles categorical encoding, date features, normalization, and leakage filtering. The model layer holds the supervised predictor and ranking logic, the explanation layer produces human-readable reasons linked to observable feature values, and the evaluation layer covers temporal holdout, predictive and ranking metrics, and robustness checks. The architecture is conceptual, and no deployed implementation is reported.

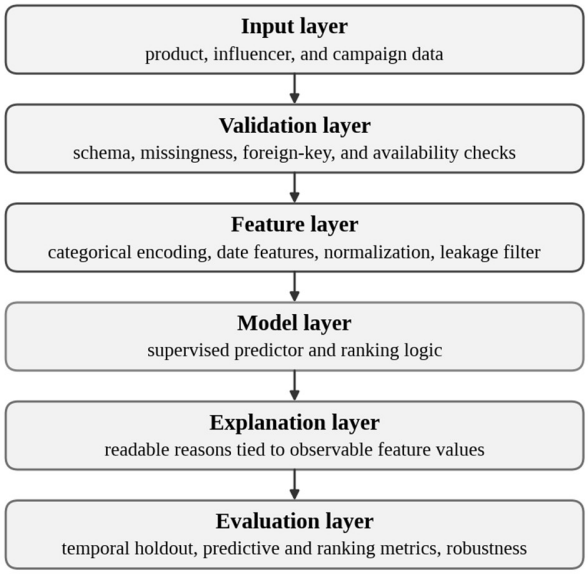



Fig. 10. Conceptual system architecture for implementation.

## XI. ETHICAL, PRIVACY, AND DATA-GOVERNANCE CONSIDERATIONS

Customer-level data are used here as source context for aggregation, not as direct ranking features. A production system should minimize personal-data exposure, apply access controls, and document the lawful grounds for processing where required. Audience location, demographic variables, and other personal attributes need particular care, because they can lead to unfair targeting or proxy discrimination. Model evaluation should therefore include subgroup checks where legally and ethically appropriate, together with an explicit policy for excluding sensitive attributes from automated ranking. If agentic AI components are later used to run or adjust campaign rankings, their roles, operational boundaries, and trust mechanisms should be defined in advance [31].

## XII. LIMITATIONS AND FUTURE WORK

### *A. Limitations*

The main analytical table has only 312 influencer–campaign–product observations, which limits model complexity and external validity. The dataset appears to be a project-supplied benchmark of Power BI style, and its independent provenance and representativeness are not established in the supplied files. Campaign-platform mismatches are frequent and may reflect differences between channel and campaign records, and platform usage patterns can themselves shift over time [26]. Audience-location match alone was a weak ranking signal, although it may still help in interaction with other factors.

The revised score was selected using training-period ranking evidence, so its advantage is dataset-specific and needs validation on independent campaigns. Some robustness cutoffs have very few test campaigns, which limits confidence in the latest estimates. The experiments do not establish causal effects of influencer selection on sales or return on investment, and revenue and order counts serve as proxy outcomes for commercial success. Finally, no completed SHAP or human-expert validation results are reported, and the original modeling source code is not available, which limits software-level reproducibility.

### *B. Future Work*

Future work should collect prospectively logged campaign data with clear decision-time snapshots and expand the number of campaigns and platforms, so that ranking estimates depend less on a single campaign mix. Actual engagement measures such as likes, comments, shares, views, and posting frequency should be added where they are available consistently across platforms. Text-based category matching from influencer bios, captions, hashtags, and product descriptions, as proposed in the conceptual framework, is a natural extension, and the TF-IDF, lexicon, and transformer-based text-classification pipelines already reported for review and social-media text [28], [29] offer starting points. Learning-to-rank models and calibrated probability models should be tested alongside random forests, fairness and subgroup performance should be evaluated for audience-related features, and blinded human-expert comparisons of model rankings should be run. Prospective or quasi-experimental campaign evaluations would move the evidence from association toward causal claims.

## XIII. CONCLUSION

This study turns the idea of influencer–product suitability into a measurable empirical workflow. Its contribution is not a claim that one universal formula can identify the best influencer. It is a disciplined way of approaching the selection problem, with explicit data lineage, feature-availability rules, temporal evaluation, ranking baselines, and interpretable outputs. On the supplied dataset, supervised models clearly outperform naive prediction, while heuristic ranking gives mixed results and depends strongly on which variables it uses.

The evidence therefore favors a hybrid design: transparent business rules for governance and explanation, combined with leakage-aware machine learning for predictive ranking. What remains uncertain is how far these results travel, since they rest on 312 observations from a project-supplied dataset and show association rather than causation. The next step is validation on larger, independently sourced, prospectively collected campaign data.

## DATA AND CODE AVAILABILITY

The empirical dataset is retained with the project research materials. The original modeling source code is not available in the retained materials, so the paper reports only implementation details supported by the documented experiment outputs and the dataset. The absence of source code limits exact software-level reproducibility and is treated as a limitation of the study.

## APPENDIX: CANDIDATE TARGET DEFINITIONS

Table XIV lists the candidate target definitions considered for the supervised experiments and the status of each.

TABLE XIV
CANDIDATE TARGET DEFINITIONS

| Candidate target | Source field | Type | Status |
|---|---|---|---|
| Future or held-out Total_Revenue | Total_Revenue | Continuous | Potential regression target; valid only when computed from held-out future campaigns |
| Future or held-out Order_Count | Order_Count | Count | Potential target for count prediction or ranking |
| Future or held-out Performance_Conversions | Performance_Conversions | Count | Potential target if the performance table is temporally aligned |
| Historical Suitability_Score | Suitability_Score | Continuous score | Descriptive ranking only; not a leakage-safe predictive target |

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